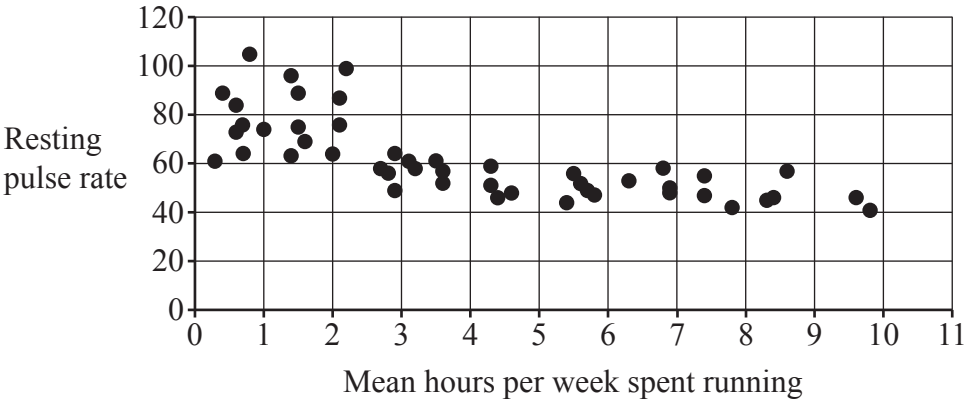
15 A personal trainer is investigating whether, in the general population, there is any association between resting pulse rate in beats per minute and mean hours per week spent running.

One week he collects a sample by asking 25 of his clients for the relevant data.

(a) (i) State the name of the sampling technique used by the personal trainer. **[1]**

(ii) Explain why this sampling technique might introduce bias. **[1]**

A biologist researching the same topic collects a random sample of size 47. She represents the data using the scatter diagram shown below.



(b) Describe the association between resting pulse rate and mean hours per week spent running. **[1]**

(c) The biologist identifies two distinct regions on the scatter diagram. Identify these regions on the copy of the scatter diagram in the **Printed Answer Booklet** by drawing an appropriate **vertical** line between them. **[1]**

According to medical research, the normal resting pulse rate for an adult is between 60 and 110 beats per minute.

The biologist separates the data into a group to the left of the vertical line on the scatter diagram and a group to the right of the vertical line on the scatter diagram. She calculates Spearman's rank correlation coefficient, r_s, and the associated p-value for each group.

The results are shown in the table.

Group	r_s	p-value
To the left of the vertical line	0.209 36	0.419 98
To the right of the vertical line	–0.613 49	0.000 31

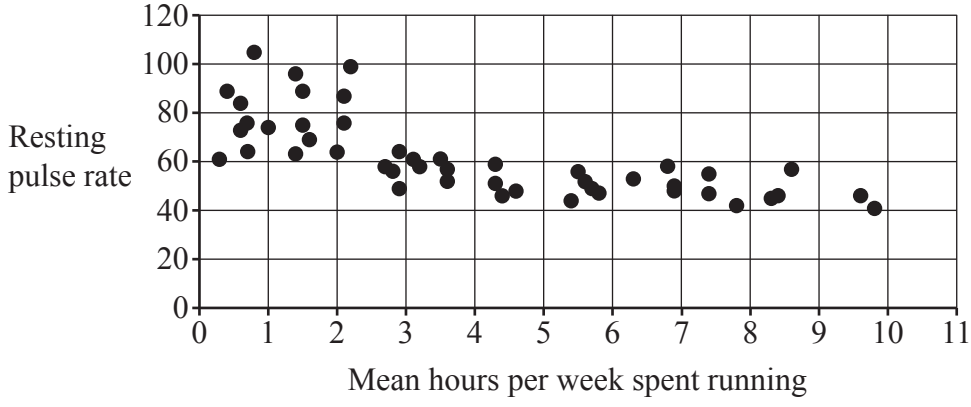
(d) With reference to the two groups identified by the biologist and to the values in the table, explain what may be inferred about the association between resting pulse rate and mean number of hours per week spent running. **[6]**

15(a)(i)

15(a)(ii)

15(b)

15(c)



15(d)	Group to the left of the vertical line $r_s = 0.20936$ p-value $= 0.41998$:
	Group to the right of the vertical line $r_s = -0.61349$ p-value $= 0.00031$: